

AMPA: Adaptive Mixed Precision Allocation for Low-Bit Integer Training

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Abstract

Low-bit integer training emerges as a promising approach to mitigate the heavy burden during network training by quantizing the weights, activations, and gradients. However, existing methods cannot well achieve mixed-precision quantization for low-bit training and are commonly limited to INT8 precision. In this paper, we propose a novel low-bit integer training framework that, for the first time, achieves adaptive mixed-precision allocation (AMPA) for weights, activations, and gradients, and pushes the boundaries to a precision level below INT8. We develop a novel magnitude-based sensitivity measurement with regard to the quantization losses of weight, activation, and gradient quantization and the average gradient magnitudes, which is demonstrated as an upper bound of quantization influence in theory. We further design a layer-wise precision update strategy under observations on the quantization losses and their effects on model performance in low-bit training. Extensive experiments on different backbones and datasets show that, compared to INT8 quantization, the proposed method can achieve more than 38% BitOPs reduction with a tolerable loss below 2% in image classification, image segmentation, and language modeling.

1. Introduction

In recent years, deep learning techniques have demonstrated remarkable progress across various tasks. Nevertheless, the success of deep neural networks (DNNs) originates from their explosive volume of model parameters. The substantial growth in computational demands and resource prerequi-

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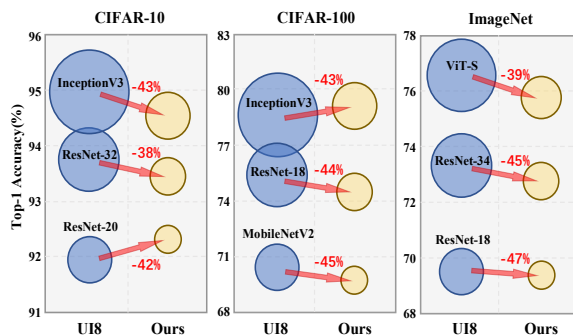


Figure 1: Image classification with the proposed AMPA low-bit training framework. For each dataset, the areas of bubbles are positively correlated with BitOPs of training the network. AMPA can save more than 38% BitOPs with tolerable loss below 2% compared with INT8 training (Zhu et al., 2020).

sites (Sevilla et al., 2022) poses unprecedented challenges to the practical deployment of deep learning methods. To mitigate this problem, network quantization has been widely studied as a promising solution that alleviates computational burdens without altering network architectures.

Network quantization can be employed in both inference and training stages (i.e., inference quantization and training quantization). Different from inference quantization that quantizes the weights or activations for acceleration in deployment (Jacob et al., 2018; Choi et al., 2018), training quantization (Cambier et al., 2020; Fu et al., 2021) simultaneously projects gradients, weights, and activations into low-precision representations during training. Since the backward computation (backpropagation) consumes nearly twice the computational cost of the forward computation (Zhao et al., 2021), low-bit training is crucial to reduce resource consumption and enable efficient training of DNNs. Compared with low-bit floating-point (FP16/FP8) training, low-bit integer training enjoys lower computation cost and better efficiency on hardware. Recent attempts (Zhu et al., 2020; Zhao et al., 2021) successfully maintain model performance under INT8 training. In this paper, we explore integer training with further lower bits and equivalent performance.

DNNs could suffer from severely degraded performance